The documentation and process conversion measures necessary to comply with this revision shall be completed by 17 June 2003.

INCH-POUND

MIL-PRF-19500/296D
17 March 2003
SUPERSEDING
MIL-PRF-19500/296C
19 September 2001

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD-EFFECT TRANSISTORS,
P-CHANNEL, SILICON TYPES 2N2609, JAN, AND UB

Inactive for new design for the 2N2609 device after 7 June 1999.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for P-channel, junction, silicon field-effect transistors. One level of product assurance is provided for each device type as specified in MIL-PRF-19500.

* 1.2 Physical dimensions. See figure 1 (similar to TO-18) and figure 2 (surface mount, UB).

* 1.3 Maximum ratings.

Types	P_T (1) T_A = +25°C	V_{GSS}	T_{STG} and T_{OP}
	mW	V dc	°C
2N2609, 2N2609UB	300	30	-65 to +200

(1) Derate linearly, 1.71 mW/°C for T_A = +25°C.

1.4 Primary electrical characteristics at T_A = +25°C.

Limit	I_{DSS} V_{DS} = -5 V dc V_{GS} = 0	$V_{GS(off)}$ V_{DS} = -5 V dc I_D = -1.0 µA dc	C_{ISS} V_{DS} = -5 V dc V_{GS} = 0 V dc f = 1 MHz	$\lvert Y_{FS} \rvert$ V_{DS} = -5 V dc V_{GS} = 0 V dc f = 1 kHz	I_{GSS} V_{GS} = 15 V dc V_{DS} = 0 V dc
	mA dc	V dc	pF	µmho	nA dc
Minimum	-2.0	0.75		2,000	
Maximum	-10.0	6.00	10	6,250	22.5

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Services (DAPS), Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

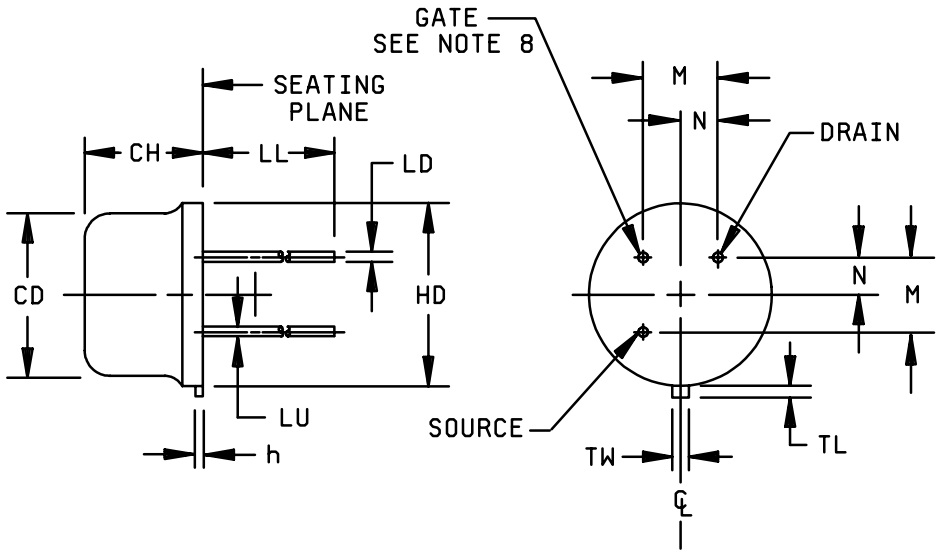
3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 and 2.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4 and table I.

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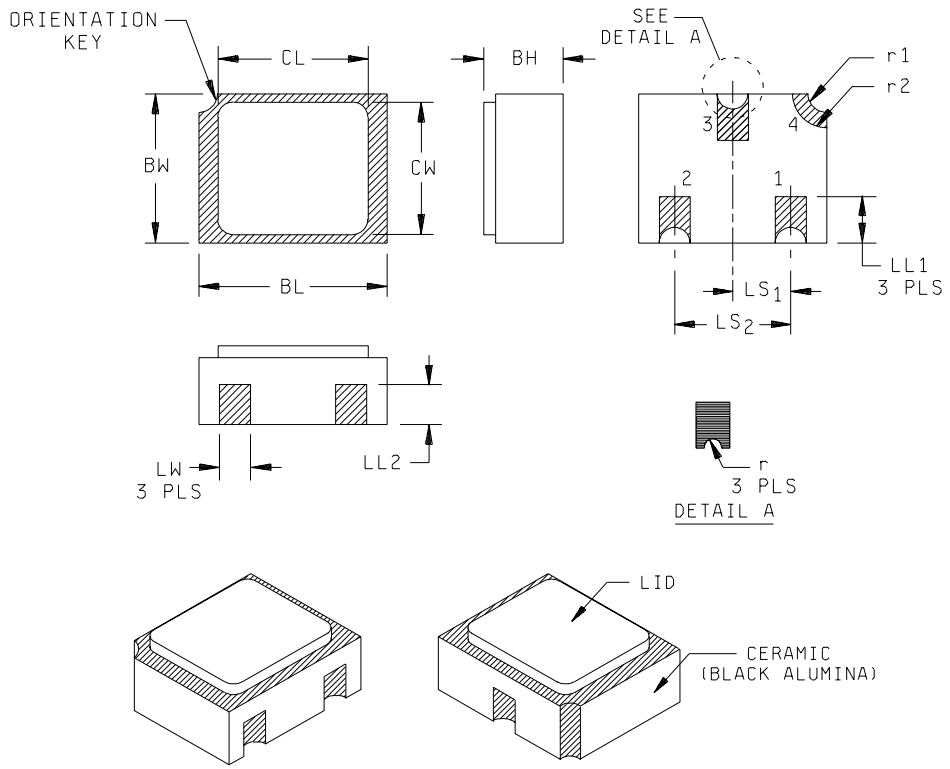


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
h		.020		0.51	
LD	.016	.021	0.41	0.53	2, 7
LL	.500	.750	12.70	19.05	7
LU	.016	.019	0.41	0.48	3,7
M	.0707 Nom		1.80 Nom		4
N	.0354 Nom		0.90 Nom		4
TL	.028	.048	0.71	1.22	6
TW	.036	.046	0.91	1.17	

NOTES:
1. Dimensions are in inches. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gauging plane .054 +.001, -.000 (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance. Figure 3 shows the preferred measurement method.
5. The gate shall be electrically connected to the case.
6. Measured from the maximum diameter of the actual device.
7. All three leads. (see 3.4.1).
8. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Physical dimensions (2N2609).

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
BH	.046	.056	0.97	1.42	
BL	.115	.128	2.82	3.25	
BW	.085	.108	2.41	2.74	
CL		.128		3.25	
CW		.108		2.74	
LL1	.022	.038	0.56	0.96	
LL2	.017	.035	0.43	0.89	
LS1	.036	.040	0.91	1.02	
LS2	.071	.079	1.81	2.01	
LW	.016	.024	0.41	0.61	
r		.008		.203	
r1		.012		.305	
r2		.022		.559	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Hatched areas on package denote metallized areas (tungsten with gold plating 60 micro inches min over 80 micro inches min nickel.
4. Lid material: Kovar; Lid finish: Gold plating 50 micro inches min thick over 80 micro inches nickel.
5. Pad 1 = Source, Pad 2 = Drain, Pad 3 = Gate, Pad 4 = Shielding connected to the lid.

* FIGURE 2. Physical dimensions, surface mount (2N2609UB).

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* 3.6 Electrical test requirements. The electrical test requirements shall be table I as specified herein.

* 3.7 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The JAN prefix can be abbreviated as J. The "2N" prefix and the "UB" suffix can also be omitted.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.3.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.3.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIb JAN of MIL-PRF-19500 and 4.3.2.1 herein. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table II herein.

4.3.2.1 Group B inspection, table VIb, of MIL-PRF-19500. Conditions for steady-state operation life are as follows: T_A = +150°C ±5°C, V_{DS} = 0; V_{GS} = 24 V dc. No heat sink or forced-air cooling on the devices shall be permitted.

4.3.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) requirements shall be in accordance with the applicable steps of table II herein.

4.3.3.1 Group C inspection, table VII, of MIL-PRF-19500. Conditions for steady-state operation life (acceleration) are as follows: 1,000 hours at V_{GS} = 24 V dc, V_{DS} = 0; T_A = +150°C ±5°C. No heat sink or forced-air cooling on the devices shall be permitted.

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TABLE I. Group A inspection .

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Gate - source breakdown voltage	3401	Bias condition C; I_G = 1.0 µA dc; V_{DS} = 0.	$V_{(BR)GSS}$	30		V dc
Gate reverse current	3411	Bias condition C; V_{GS} = 30 V dc; V_{DS} = 0.	I_{GSS1}		30	nA dc
Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0.	I_{GSS2}		22.5	nA dc
Zero-gate-voltage drain current	3413	Bias condition C; V_{DS} = 5 V dc; V_{GS} = 0.	I_{DDSS1}	-2.0	-10.0	mA dc
Gate to source cutoff voltage	3403	I_D = 1 µA dc; V_{DS} = 5 V dc	$V_{GS(off)}$	0.75	6.0	V dc
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Drain current	3413	Bias condition C, V_{GS} = 0 V dc; V_{DS} = 5	I_{DDSS2}	-1.0		mA dc
Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0.	I_{GSS3}		45	µA dc
Low-temperature operation:		T_A = -55°C				
Small-signal, common-source, short-circuit, forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 kHz.	$\lvert Y_{fs1} \rvert$		9,375	µmho

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal common-source, short-circuit, forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 kHz.	$\lvert Y_{fs2} \rvert$	2,000	6,250	µmho
Small-signal common-source, short-circuit, input capacitance	3431	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 MHz.	C_{iss}		10	pF
Common-source spot noise figure		V_{DS} = 5 V dc; V_{GS} = 0; B_W = 16 percent; R_G = 1 Mohm; e_{gen} = 1.82 mV; R_L = 220 ohms; f = 1 kHz; (see figure 4).	NF		3	dB
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups A, B, and C electrical measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1	Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS1}		23	nA dc
2	Gate reverse current	3411	Bias condition C; V_{GS} = 15 V dc; V_{DS} = 0	I_{GSS2}		45	nA dc
3	Drain current	3413	Bias condition C; V_{GS} = 5 V dc; V_{DS} = 0	I_{DSS1}	-2.0	-10.0	mA dc
4	Drain current	3413	Bias condition C; V_{GS} = 5 V dc; V_{DS} = 0	I_{DSS2}	-1.80	-11.0	mA dc
5	Small-signal, common-source short-circuit, forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 kHz	$\lvert Y_{fs1} \rvert$	2,000	6,250	µmho
6	Small-signal, common-source short-circuit, forward transfer admittance	3455	V_{DS} = 5 V dc; V_{GS} = 0; f = 1 kHz	$\lvert Y_{fs3} \rvert$	1,800	6,875	µmho

* 1/ The electrical measurements for table VIb (JAN) of MIL-PRF-19500 are as follows:
  a. Subgroup 2, see table II herein, steps 1, 3 and 5.
  b. Subgroups 3 and 6, see table II herein, steps 2, 4 and 6.

2/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
  a. Subgroups 2 and 3, see table II herein, steps 1, 3 and 5.
  b. Subgroup 6, see table II herein, steps 2, 4 and 6.

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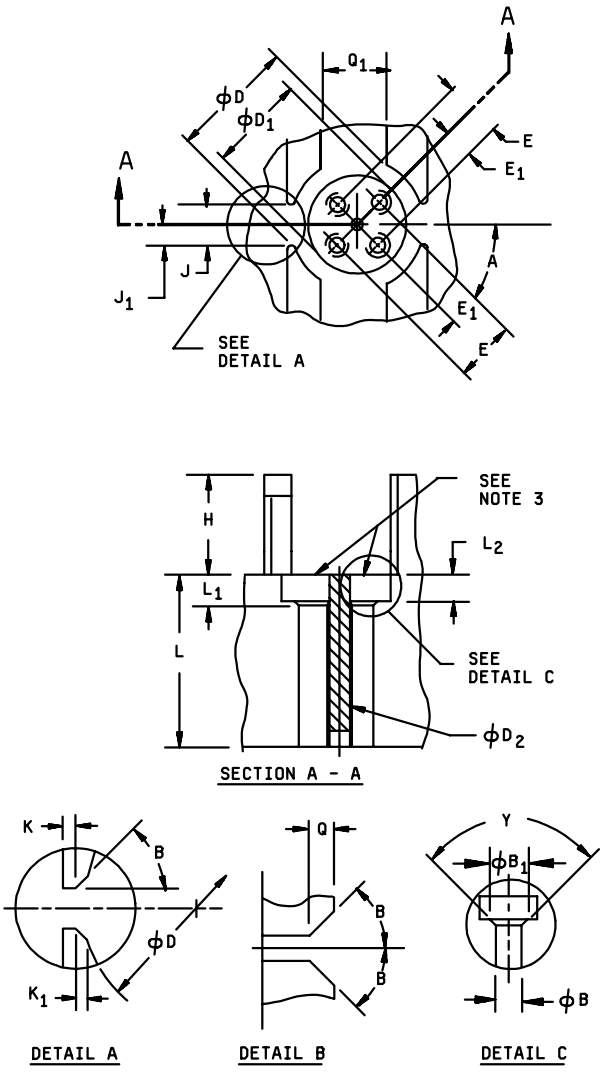


FIGURE 3. Gauge for lead and tab location for transistor type 2N2609.

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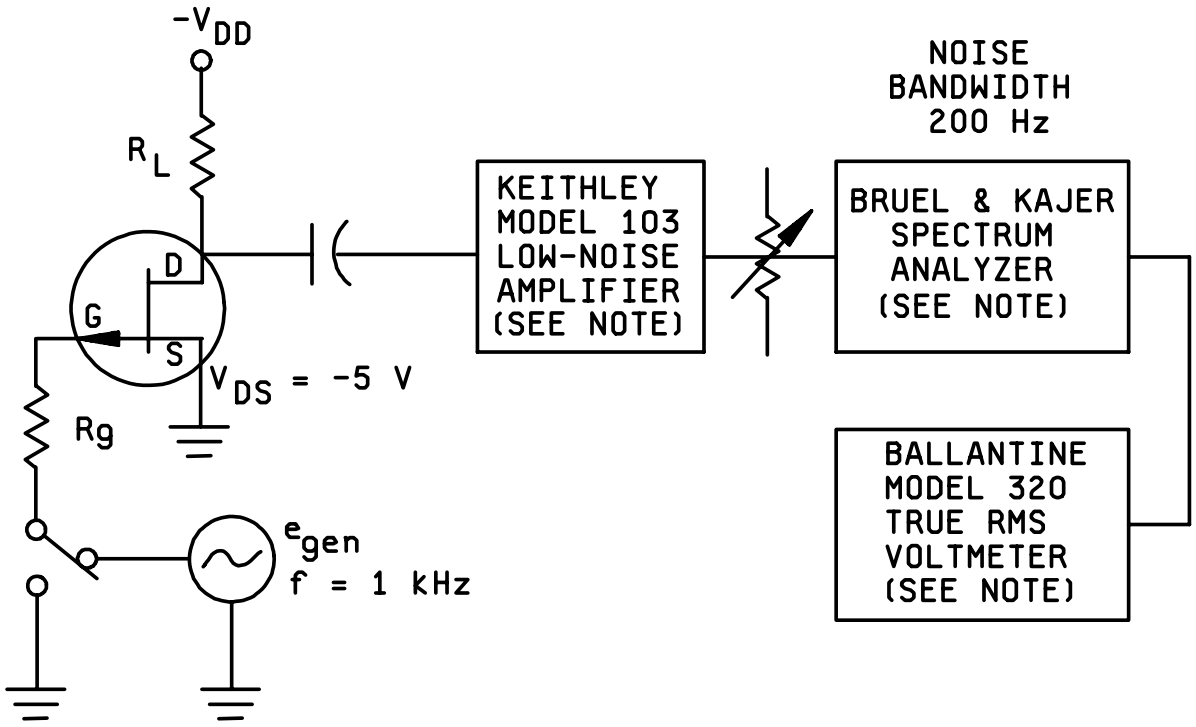
Symbols	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.0325	.0335	0.83	0.85	4
ϕB_1	.043 Nom		1.09 Nom		4
ϕD	.2310	.2315	5.87	5.88	
ϕD_1	.159	.161	4.04	4.09	
ϕD_2	.040 Nom		1.02 Nom		5
E	.0995	.1005	2.53	2.55	
E_1	.0495	.0505	1.26	1.28	
H	.145	.155	3.68	3.94	
J	.0470	.0475	1.19	1.21	
J_1	.0235	.0245	0.60	0.62	
K	.009	.011	0.23	0.28	
K_1	.005Nom		.127 Nom		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
L_2	.043 Nom		1.09 Nom		
Q	.040 Nom		1.02 Nom		
Q_1	.123	.127	3.12	3.23	
A	44.90°	45.10°			
B	45° Nom				
Y	90° Nom				

NOTES:

1. Dimensions are in inches. Metric equivalents are given for general information only.
2. The following gauge procedure shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) +.010 inch (0.25 mm) from seating surface of the gauge. A force of 8 ±.5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not to be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
3. These surfaces to be parallel and in same plane within ±.001 inch (0.03 mm).
4. Four holes.
5. Pressed in.

FIGURE 3. Gauge for lead and tab location for transistor type 2N2609 - Continued.

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NOTE: Or equivalents

Procedure: The voltage and current shall be applied to the terminals, and the noise figure shall then be measured as follows:

1. Connect egen to the input of the device under test.
2. Adjust gain of system to give 0 dB reading on a convenient scale of the Ballintine voltmeter or equivalent.
3. Switch the device input to ground.
4. Increase system gain by 60 dB.
5. Read noise figure directly in dB's on Ballintine scale or equivalent.

FIGURE 4. Common-source spot noise figure test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Packaging requirements (see 5.1).

d. Lead finish (see 3.4.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers' List (QML) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - MI, SM
Navy - AS, MC, OS, SH
Air Force - 19, 99

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/296D	2. DOCUMENT DATE 17 March 2003

3. **DOCUMENT TITLE** SEMICONDUCTOR DEVICE, FIELD-EFFECT TRANSISTORS, P-CHANNEL, SILICON, TYPES 2N2609, JAN, AND UB.

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan.barone@dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

DD Form 1426, Feb 1999 (EG) Previous editions are obsolete WHS/DIOR, Feb 99

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